

The molecular neuropathology of amyotrophic lateral sclerosis: protocol for systematic review and meta-analysis

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Keywords

ALS, FTD, neuropathology, systematic review, meta-analysis.

Introduction

Amyotrophic lateral sclerosis (ALS) is a phenotypically heterogeneous multi-system neurodegenerative condition caused by the selective death of motor neurones. The resultant progressive muscle atrophy leads to respiratory failure and a median survival duration of around two years; however, 10% of patients may survive for more than 10 years (1, 2). Significant variation amongst the clinical features has led to the recognition of phenotypes defined by the relative degree of upper or lower motor neuron involvement, bulbar impairment or deficits in cognition and language (3-6). Even within these subtypes, there is further variation across both the severity of the symptoms and median survival duration. The emerging genetic background is also highly heterogeneous: over 30 ALS associated genes have now been identified and the list is increasing (7). Only 10% of cases exhibit monogenic heritability and are described as familial, whilst the remainder are likely to be due to complex interactions of multiple latent genetic alterations (8). Critically, the examination of human pathological tissue specimens has confirmed the link between ALS and Frontotemporal dementia (FTD), which as TDP-43 proteinopathies are now understood to exist on an ALS-FTD clinico-pathological spectrum (9,10). The identification of a limited range of protein aggregates as pathological hallmarks unifies both clinically and genetically distinct ALS cases, suggesting convergence of key degenerative pathways.

A significant body of neuropathological research examining the molecular heterogeneity across ALS phenotypes exists and an up to date review of these studies is now required. Through conducting a systematic review and meta-analysis of the existing ALS human neuropathology literature we aim to outline the current pathological molecular profile of ALS. This pathological insight is critical to understanding the disease pathophysiology and developing targeted therapeutics for specific molecular pathways, applicable to people who have different ALS phenotypes. Human pathological studies to date have revolutionised our understanding of the ALS disease aetiology. Here we will formally assess these studies to identify the key factors and pathways implicated in this devastating condition.

Approach

A systematic review will be performed assessing quantitative molecular assays performed on human ALS and ALS-FTD tissue compared to controls. Individual meta-analyses will be performed for each of the quantitative intervention methods. A methodological quality assessment will be performed with individual study quality checklist scores.

Objectives

PICOS framework

Population: Neuropathological studies of brain or spinal cord samples from patients with ALS and ALS-FTD.

Intervention: All quantitative molecular assays (*northern blot, qRT-PCR and RNA-Seq, molecular bar coding, western blot, ELISA, mass spectrometry, in-situ hybridisation/ immunofluorescence/ immunohistochemistry.*)

Comparison: Control neuropathological samples (healthy or non-neurological conditions or non-ALS)

Outcome measure

Primary outcome: differential molecular species quantification.

Secondary outcomes: assessment of study heterogeneity.

Further subgroup analyses of non-comparative pathological features;

Hypothesis 1 - *Dysregulation of common pathways is associated with clinical features i.e. the upregulation of ER stress mediators will correlate with affected regions/samples and these pathways may be conserved across different neurodegenerative diseases.*

- List of most dysregulated genes and pathway analysis/ (clinical overlap with other neurodegenerative disease/disease co-occurrence).
- List of pathological correlates of clinical features such as long and short survival and cognitive status (continuous/ordinal data to be displayed in a forest plot and categorical/qualitative data in a table).

Hypothesis 2 - *The severity/burden of pathological features may not correlate with clinical features due to cell type specific effects.*

- The association of; ordinal proteinopathy severity grading, regional burden across studies (proportion of mild, moderate and severe grade) and qualitative regional pathological features with clinical correlates of; genetic status, age at onset, disease duration, cognitive status (using a validated measurement scale), site of onset/regional spread.

Hypothesis 3 - *The presence of environmental neurotoxin β -N-methylamino-L-alanine (L-BMAA) correlates with disease status and or clinical parameters.*

- Identification of L-BMAA in ALS cases and correlation with phenotypic features.

Hypothesis 4 - *The presence of viral elements correlate with disease status and or clinical parameters.*

- The association of viruses (including endogenous viral elements) with clinical parameters in ALS cases.

Study design

All studies quantifying the molecular features of ALS neuropathology samples compared to controls.

Methods

Search methods

Sources: databases: 1. PubMed, 2. MEDLINE, 3. EMBASE

Search date: 07/06/2019

There will be no publication date restrictions or language restrictions

Search terms;

PubMed;

((“human” AND “post-mortem” OR “PM” OR “post mortem”)) AND (“ALS” or “MND”)

MEDLINE;

(human and (post-mortem or PM or post mortem) and (ALS or MND))

EMBASE;

(human and (post-mortem or PM or post mortem) and (ALS or MND))

Screening

The Systematic Review Facility online screening tool (app.syrf.org.uk) will be used by two independent authors to screen the titles and abstracts based on predefined criteria. In the event of discordance between authors (concordance <0.66), a consensus decision will be made. Selected studies will be imported to EndNote and duplicates discarded. Study quality will be assessed by one reviewer against the CAMARADES' study quality checklist (adapted) and data will be extracted by one reviewer as below. Additional study quality assessment and data extraction will be carried out by an independent reviewer for 10% of the selected studies and inter-observer variability assessed.

Eligibility

Inclusion criteria;

All quantitative molecular studies of human ALS neuropathological tissue compared to a control group. Studies that do not include a control group will be assessed for inclusion on an individual basis.

Exclusion criteria;

Animal studies
Reviews
Letters and comments
Case reports
Neuroimaging studies

Quality checklist

CAMARADES' study checklist- adapted as follows;

8 items will be considered, the median number of checklist items scored and the interquartile range will be calculated.

1. Peer reviewed publication
2. Blinded assessments
3. Sample size calculation
4. Sample clinical and genetic profiling
5. Statement of age/gender matched controls selected
6. Statement regarding quality of tissue including but not limited to; Post-mortem interval/RNA integrity number (RIN)/pH
7. Appropriate statistical methods used
8. Statement of potential conflict of interest

Study characteristics to be extracted;

- Study citation information i) author ii) year
- Sample size disease: controls
- Study sample clinical profile; disease diagnosis and duration (ALS or ALS-FTD)
- Genetic status of ALS samples
- Disease/clinical profile of controls (healthy, other)
- Sex of patients and controls
- Mean age of patients and controls
- Entity detected; DNA, mRNA, MiRNA, lncRNA, protein
- Intervention quantification assay; outcome type and value
- Regions assessed (BA or specific neuroanatomical regions)
- Key findings
- All outcome measures for patients and controls (increase/decrease value)
- Markers of tissue quality: e.g. post mortem interval; pH; RIN

Statistical analysis

Given the heterogeneity of intervention quantification methods assessed, standardised mean differences (SMD) will be used to describe outcome measures to enable inter-study comparison.

SMD will be compared using Hedges' g-statistic, to account for bias from small sample sizes, using a random effects model. SMDs will be reported as odds ratios with 95% confidence intervals. Heterogeneity will be assessed for all outcome measures using I²-values, a funnel plot and Egger's regression test will be used to assess publication bias.

Conflicts of interest

The authors have no conflicts of interest.

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